

ZORK III

The Dungeon Master — A Concise Full-Potential Walkthrough, with an Opcode Census

Infocom, 1982 (`stories/zork3.z3`, Z-machine version 3), played on the zAI-Machine interpreter with random seed 42. Final result: **potential 7 of a possible 7, in 249 moves**, with no deaths. The story ends in the Treasury of Zork, where you take the Dungeon Master's place. Every command below is replayed by `ZorkIIWalkthroughTests.ZorkIII_FullPotential`, which also tallies each opcode executed (see the Opcode Census at the end).

Solved during play with the game's manual as the only outside reference. No published walkthrough or hints were consulted. The interpreter's disassembler and object/global inspection tooling were used to understand puzzles. That is how the Dungeon Master's seven-item test, the ladder exit from the Royal Puzzle, and the light needed in the Damp Passage were found.

Know Before You Go

- **There is no treasure score.** Zork III rates your *potential*: seven points, one for each trial of character. They are your first visit to the Cliff Ledge, the sword appearing, fighting the hooded figure, entering the lake, your first trip by the Scenic Vista table, your first time-machine trip, and your first push in the Royal Puzzle.
- **The Dungeon Master checks your bag.** When you knock at the Dungeon Entrance, he admits you only if you carry seven things: the staff, amulet, hood, cloak, ancient book, strange key and ring. Otherwise he sends you off with a hint. So the book must leave the Royal Puzzle by the ladder, not through the steel door's slot, which confiscates it.
- **The earthquake is on a timer.** It strikes at a fixed turn chosen at the start of the game. It opens the cleft into the Museum, but it also breaks the aqueduct, which is the only way back from the Key Room. Get the key first.
- **The lake takes everything.** Swimming in makes you drop all you carry. The lamp dies, the torch goes out and the bread is lost. Leave your gear on the shore.
- **Kindness counts.** Feed the old man before you fight. The hooded figure is the same person, and after the fight the old man is gone. Defeat the figure, but don't kill him.
- **Death is survivable twice.** You wake at the Endless Stair, but your belongings are scattered. This route never dies.

The Walkthrough

1. The Old Man (+0)

Command	What happens / why	Pts
VERBOSE, TAKE LAMP, TURN ON LAMP, S, W, W, TAKE BREAD, E, E, E, NE	Bread from the Shore, then on to the Engravings Room.	
WAKE MAN, GIVE BREAD TO MAN, HELLO	He eats, rests against the wall, and gives away the secret door. The HELLO only advances the random number stream to set up a winning fight.	

2. The Hooded Figure (+2)

Command	What happens / why	Pts
SW, W, W, SW	In the Land of Shadow a sword drops at your feet.	+1
ATTACK FIGURE WITH SWORD ×13	Keep swinging until he is badly hurt and defenseless. Then stop; he is the old man.	+1
TAKE HOOD, TAKE CLOAK		

3. Trust at the Cliff (+1)

Command	What happens / why	Pts
N, W, D	The Cliff Ledge, where a heavy chest sits.	+1
TIE ROPE TO CHEST, WAIT ×5, GRAB ROPE	Your "friend" at the top hauls the chest up and leaves. He returns, pulls you up, and hands you a plain wooden staff from the bottom of the chest.	

4. The Flathead Ocean (+0)

Command	What happens / why	Pts
D, D, S, WAIT ×3, HELLO SAILOR, TAKE VIAL	A boat sails past. The words he has waited three ages to hear bring it to shore, with a vial of invisible liquid.	
N, NE, SE	Back toward the Junction.	

5. The Lake (+1)

Command	What happens / why	Pts
E, S, S, S, HELLO, HELLO, DROP ALL	At the Lake Shore. The two HELLOs set up a successful grab.	
ENTER LAKE, D, TAKE OBJECT, U, N, TAKE ALL	Dive for the golden amulet and wear it.	+1

6. The Scenic Vista (+1)

The table's indicator cycles every two turns: I (Timber Room), II (Room 8), III (Damp Passage), IV (Sacrificial Altar, which is deadly). Touching the table sends you there until the indicator changes.

Command	What happens / why	Pts
DROP ALL, ENTER LAKE, W, S, TAKE TORCH	Swim to the Western Shore and walk to the Vista.	
TOUCH TABLE, TAKE REPELLENT, WAIT	Room 8 holds a can of Frobozz Magic Grue Repellent.	+1
TOUCH TABLE, DROP TORCH, WAIT, N	Leave the lit torch in the Damp Passage. It will light your way back.	

7. The Key Room (+0)

Command	What happens / why	Pts
SPRAY REPELLENT ON ME, ENTER LAKE, S, S, S, E, TAKE KEY	The repellent lasts just long enough to cross the Dark Places to the Key Room.	
MOVE COVER, D, N, N, D	Down the manhole into the aqueduct, across the High Arch, and down the water slide to the Damp Passage.	
TAKE TORCH, W, S, S, S, TAKE ALL	Back to the Lake Shore for your gear.	

8. The Earthquake (+0)

Command	What happens / why	Pts
N, N, E, E, S, S, WAIT	To the Great Door. The quake cracks open a cleft beside it.	

9. The Time Machine (+1)

Command	What happens / why	Pts
E, OPEN STONE DOOR, DROP ALL, N, PUSH GOLD MACHINE SOUTH, PUSH GOLD MACHINE EAST	Wheel the machine from the Technology Museum into the Jewel Room.	
SIT ON SEAT, TURN DIAL TO 776, PUSH BUTTON, TAKE RING, WAIT ×2	Back to the year 776, while the jewels are still on display. Wait for the guards to march away.	+1
OPEN DOOR, W, OPEN WOODEN DOOR, N, PUT RING UNDER SEAT, SIT ON SEAT, TURN DIAL TO 948, PUSH BUTTON, GET UP, LOOK UNDER SEAT	The ring stays under the machine's seat while time passes.	
OPEN DOOR, S, TAKE ALL		

10. The Royal Puzzle (+1)

A 6×6 maze of sliding sandstone walls. PUSH <dir> WALL slides a wall and you follow it. The book is in a niche, and one wall has a ladder on it. Climb out by moving the ladder beside the entrance square.

Command	What happens / why	Pts
S, D, PUSH SOUTH WALL, N, PUSH EAST WALL, S, S, E, E, N, N, PUSH EAST WALL, W, S, S, PUSH SOUTH WALL, TAKE BOOK	<i>The first push earns the point.</i>	+1
PUSH SOUTH WALL, E, E, N, N, N, PUSH WEST WALL, N, W, PUSH SOUTH WALL ×3, PUSH EAST WALL, S, S, PUSH WEST WALL, PUSH NORTH WALL, E, N, PUSH WEST WALL ×2, E, S, PUSH WEST WALL ×2, PUSH NORTH WALL ×3, W, N, UP, N	<i>Found by breadth-first search over the wall layout. It brings the ladder wall next to the hole you came in by.</i>	

11. The Secret Door and the Mirror (+0)

Command	What happens / why	Pts
W, N, N, W, W, N, E, NE, OPEN DOOR, N	<i>Back to the Engravings Room and down the secret stairs to the Button Room.</i>	
N, DROP SWORD, S, PUSH BUTTON, N, N, N	<i>The sword blocks the red beam. Then the button opens the mirror in the Hallway.</i>	

12. Inside the Mirror (+0)

The box turns when you push the coloured panels (while the short pole is raised) and slides along a channel when you push the mahogany panel. Leave through the pine panel only when it faces north; elsewhere the Guardians of Zork strike.

Command	What happens / why	Pts
RAISE SHORT POLE, PUSH RED PANEL ×2, LOWER SHORT POLE, PUSH MAHOGANY PANEL ×3	<i>Face north and slide to the end of the channel.</i>	
RAISE SHORT POLE, PUSH RED PANEL ×4, LOWER SHORT POLE, PUSH PINE PANEL, N	<i>Turn to face south so the pine panel opens north, onto the Dungeon Entrance.</i>	

13. The Dungeon Master (+0)

Command	What happens / why	Pts
KNOCK ON DOOR	<i>Carrying all seven items, you are admitted, and the Dungeon Master follows you.</i>	
N, W, N, N, TURN DIAL TO 4, PUSH BUTTON, DUNGEON MASTER, STAY	<i>At the Parapet, the sundial brings cell 4, with its bronze door, round to the cell door.</i>	
S, OPEN DOOR, S, DUNGEON MASTER, TURN DIAL TO 1, DUNGEON MASTER, PUSH BUTTON	<i>With you inside, he rotates the cell away from the corridor.</i>	
UNLOCK BRONZE DOOR WITH KEY, OPEN BRONZE DOOR, S	<i>The Treasury of Zork, where you become the new Dungeon Master.</i>	

Where the Potential Comes From

Each point is awarded once, the first time the deed is done. Move numbers refer to the full command list.

Move	Command	Potential point
18	SW (Land of Shadow)	The sword appears: the figure accepts your challenge
19	ATTACK FIGURE WITH SWORD	First blow against the hooded figure
36	D (Cliff Ledge)	Reaching the ledge, where trust is tested
62	ENTER LAKE	Braving the icy lake
73	TOUCH TABLE	First journey by the Scenic Vista table
113	PUSH BUTTON (dial 776)	First trip in the time machine
132	PUSH SOUTH WALL	First move in the Royal Puzzle

Opcode Census

Every instruction executed during the 249-move run was decoded with the interpreter's own disassembler and counted by mnemonic. The run used **53 distinct opcodes** over **200,686** instructions. As a version 3 story, Zork III uses the classic call (`call_vs` here), `read` (`sread`) and the stack opcodes `push/pull`. It ends with `quit` after the Treasury.

#	Opcode	Form:No.	Executions	Purpose
1	<code>jz</code>	1OP:128	35,248	branch if zero
2	<code>je</code>	2OP:1	29,487	branch if equal to any operand
3	<code>loadw</code>	2OP:15	16,798	load word from array
4	<code>jump</code>	1OP:140	15,959	unconditional jump
5	<code>store</code>	2OP:13	13,316	store to variable
6	<code>call_vs</code>	VAR:224	12,919	call routine (V3 "call"), store result
7	<code>add</code>	2OP:20	9,529	signed addition
8	<code>storew</code>	VAR:225	5,590	store word into array
9	<code>inc_chk</code>	2OP:5	5,455	increment and branch if greater
10	<code>test_attr</code>	2OP:10	5,225	branch if object has attribute
11	<code>loadb</code>	2OP:16	5,080	load byte from array
12	<code>get_prop_addr</code>	2OP:18	3,812	address of property data
13	<code>jg</code>	2OP:3	3,217	branch if greater (signed)
14	<code>jl</code>	2OP:2	3,115	branch if less (signed)
15	<code>sub</code>	2OP:21	3,087	signed subtraction
16	<code>test</code>	2OP:7	2,820	branch if all bitmap flags set
17	<code>get_child</code>	1OP:130	2,685	first child object, branch if any
18	<code>get_sibling</code>	1OP:129	2,505	next sibling, branch if any
19	<code>get_prop_len</code>	1OP:132	2,188	length of property data
20	<code>get_parent</code>	1OP:131	2,064	parent object
21	<code>get_prop</code>	2OP:17	1,858	read property value
22	<code>ret_popped</code>	0OP:184	1,749	return top of stack
23	<code>ret</code>	1OP:139	1,718	return value
24	<code>rfalse</code>	0OP:177	1,613	return false
25	<code>rtrue</code>	0OP:176	1,581	return true
26	<code>push</code>	VAR:232	1,472	push onto stack
27	<code>print</code>	0OP:178	1,415	print inline string
28	<code>div</code>	2OP:23	1,378	signed division

#	Opcode	Form:No.	Executions	Purpose
29	dec_chk	2OP:4	1,048	decrement and branch if less
30	and	2OP:9	967	bitwise AND
31	inc	1OP:133	849	increment variable
32	new_line	0OP:187	773	print newline
33	get_next_prop	2OP:19	731	next property number of an object
34	mul	2OP:22	677	signed multiplication
35	mod	2OP:24	496	signed remainder
36	jin	2OP:6	457	branch if object is a child of
37	pull	VAR:233	264	pop stack into variable
38	storeb	VAR:226	247	store byte into array
39	print_obj	1OP:138	227	print object short name
40	read	VAR:228	226	read a line of input (V3 sread)
41	insert_obj	2OP:14	193	move object into another
42	set_attr	2OP:11	142	set object attribute
43	print_paddr	1OP:141	121	print string at packed address
44	random	VAR:231	117	random number / reseed
45	print_ret	0OP:179	113	print inline string, newline, return true
46	clear_attr	2OP:12	68	clear object attribute
47	or	2OP:8	45	bitwise OR
48	remove_obj	1OP:137	11	detach object from parent
49	print_num	VAR:230	10	print signed number
50	dec	1OP:134	7	decrement variable
51	load	1OP:142	7	load variable value (no pop)
52	print_char	VAR:229	6	print ZSCII character
53	quit	0OP:186	1	end the game

Full Command List

One line per input, exactly as replayed by the test (seed 42). Read down each column.

1 verbose	77 drop torch	153 push west wall
2 take lamp	78 wait	154 north
3 turn on lamp	79 north	155 west
4 south	80 spray repellent on me	156 push south wall
5 west	81 enter lake	157 push south wall
6 west	82 south	158 push south wall
7 take bread	83 south	159 push east wall
8 east	84 south	160 south
9 east	85 east	161 south
10 east	86 take key	162 push west wall
11 northeast	87 move cover	163 push north wall
12 wake man	88 down	164 east
13 give bread to man	89 north	165 north
14 hello	90 north	166 push west wall
15 southwest	91 down	167 push west wall
16 west	92 take torch	168 east
17 west	93 west	169 south
18 southwest	94 south	170 push west wall
19 attack figure with sword	95 south	171 push wall
20 attack figure with sword	96 south	172 push north wall
21 attack figure with sword	97 take all	173 push north wall
22 attack figure with sword	98 north	174 push north wall
23 attack figure with sword	99 north	175 west
24 attack figure with sword	100 east	176 north
25 attack figure with sword	101 east	177 up
26 attack figure with sword	102 south	178 north
27 attack figure with sword	103 south	179 west
28 attack figure with sword	104 wait	180 north
29 attack figure with sword	105 east	181 north
30 attack figure with sword	106 open stone door	182 west
31 attack figure with sword	107 drop all	183 west
32 take hood	108 north	184 north
33 take cloak	109 push gold machine south	185 east
34 north	110 push gold machine east	186 northeast
35 west	111 sit on seat	187 open door
36 down	112 turn dial to 776	188 north
37 tie rope to chest	113 push button	189 north
38 wait	114 take ring	190 drop sword
39 wait	115 wait	191 south
40 wait	116 wait	192 push button
41 wait	117 open door	193 north
42 wait	118 west	194 north
43 grab rope	119 open wooden door	195 north
44 down	120 north	196 raise short pole
45 down	121 put ring under seat	197 push red panel
46 south	122 sit on seat	198 push red panel
47 wait	123 turn dial to 948	199 lower short pole
48 wait	124 push button	200 push mahogany panel
49 wait	125 get up	201 push mahogany panel
50 hello sailor	126 look under seat	202 push mahogany panel
51 take vial	127 open door	203 raise short pole
52 north	128 south	204 push red panel
53 northeast	129 take all	205 push red panel
54 southeast	130 south	206 push red panel
55 east	131 down	207 push red panel
56 south	132 push south wall	208 lower short pole
57 south	133 north	209 push pine panel
58 south	134 push east wall	210 north
59 hello	135 south	211 knock on door
60 hello	136 south	212 north
61 drop all	137 east	213 west
62 enter lake	138 east	214 north
63 down	139 north	215 north
64 take object	140 north	216 turn dial to 4
65 up	141 push east wall	217 push button
66 north	142 west	218 dungeon master, stay
67 take all	143 south	219 south
68 drop all	144 south	220 open door
69 enter lake	145 push south wall	221 south
70 west	146 take book	222 dungeon master, turn dial to 1
71 south	147 push south wall	223 dungeon master, push button
72 take torch	148 east	224 unlock bronze door with key
73 touch table	149 east	225 open bronze door
74 take repellent	150 north	226 south
75 wait	151 north	
76 touch table	152 north	